

Lab 3

Aim: To design and represent an Entity Relationship (ER) diagram for a real-world database system using the draw.io tool by identifying entities, attributes, keys, relationships and cardinalities.

Integrated University Research Management System

Design a complete ER model for a University Research Management System. The system should maintain information about Researchers, Departments, Research Projects, Students, Publications, and Project Dependents/Assistants.

A Researcher has a unique Researcher_ID, Name, Date_of_Birth, Email, multiple Phone Numbers, and Address. Address consists of Street, City, State, and PIN Code. Age is calculated from Date_of_Birth.

Each researcher belongs to a Department. A department has Department_ID, Department_Name, and Location.

A researcher can work on multiple Research Projects. Each project has Project_ID, Project_Title, Start_Date, End_Date, Budget, and multiple Keywords. A project may involve multiple researchers.

A researcher can have multiple Students working under them. Each student has Student_ID, Name, Email, and Date_of_Birth.

Each research project can produce multiple Publications. A publication has Publication_ID, Title, Publication_Date, DOI, and Publisher.

A project may also have Project Assistants. An assistant is identified only within the context of its project and has Assistant_Name, Role, and Joining_Date. The same assistant name may occur in different projects.

Identify all entity types and entity sets.

Researcher-Represents all researchers working in the university.

Department-Represents all departments in the university.

Research Project-Represents all research projects undertaken by the university.

Student-Represents all students working under researchers.

Publication-Represents all publications produced from research projects.

Project Assistant-Represents all assistants associated with research projects.

2. Identify strong and weak entity sets.

Strong and Weak Entity Sets

- **Strong Entity Sets:**
- Researcher
- Department
- Research Project
- Student
- Publication
- **Weak Entity Set:**
- Project Assistant

3. Identify the owner entity and partial key of the weak entity.

Weak Entity: Project Assistant

Owner Entity: Research Project

Partial Key: Assistant_Name

4. Identify simple attributes.

The simple attributes in the University Research Management System are:

- **Researcher:** Researcher_ID, Date_of_Birth, Email
- **Department:** Department_ID, Department_Name, Location
- **Research Project:** Project_ID, Project_Title, Start_Date, End_Date, Budget
- **Student:** Student_ID, Name, Email, Date_of_Birth
- **Publication:** Publication_ID, Title, Publication_Date, DOI, Publisher
- **Project Assistant:** Assistant_Name, Role, Joining_Date

5. Identify composite attributes.

The composite attributes in the system are:

1. **Researcher** → **Name**
 - First_Name
 - Middle_Name
 - Last_Name
2. **Researcher** → **Address**
 - Street
 - City
 - State
 - PIN_Code

6. Identify single-valued attributes.

Researcher

- Researcher_ID
- Name (composite, but single-valued)
- Date_of_Birth
- Email
- Address (composite, but single-valued)

Department

- Department_ID
- Department_Name
- Location

Research Project

- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget

Student

- Student_ID
- Name

- Email
- Date_of_Birth

Publication

- Publication_ID
- Title
- Publication_Date
- DOI
- Publisher

Project Assistant

- Assistant_Name
- Role
- Joining_Date

7. Identify multi-valued attributes

The multi-valued attributes in the system are:

1. **Researcher → Phone_Number**
A researcher can have multiple phone numbers.
2. **Research Project → Keyword**
A research project can have multiple keywords.

8. Identify stored attributes.

Researcher

- Researcher_ID
- Name (First_Name, Middle_Name, Last_Name)
- Date_of_Birth
- Email
- Phone_Number
- Address (Street, City, State, PIN_Code)

Department

- Department_ID
- Department_Name
- Location

Research Project

- Project_ID
- Project_Title
- Start_Date
- End_Date
- Budget
- Keyword

Student

- Student_ID
- Name
- Email
- Date_of_Birth

Publication

- Publication_ID
- Title
- Publication_Date
- DOI
- Publisher

Project Assistant

- Assistant_Name
- Role
- Joining_Date

9. Identify derived attributes.

Derived Attribute- Age

Age is derived from Date_of_Birth.

10. Identify key attributes.

Entity	Key Attribute	Key Status
Researcher	Researcher_ID	Primary Key
Department	Department_ID	Primary Key
Research Project	Project_ID	Primary Key
Student	Student_ID	Primary Key
Publication	Publication_ID	Primary Key

Project Assistant	Assistant_Name	Partial Key
Project Assistant	Project_ID + Assistant_Name	Composite Key

11. Identify candidate keys and select appropriate primary keys.

Entity	Candidate Key	Selected Primary Key
Researcher	Researcher_ID	Researcher_ID
Department	Department_ID	Department_ID
Research Project	Project_ID	Project_ID
Student	Student_ID	Student_ID
Publication	Publication_ID	Publication_ID
Project Assistant	(Project_ID,Assistant_Name)	(Project_ID,Assistant_Name)

12. Identify at least one composite key.

Assitant_Name

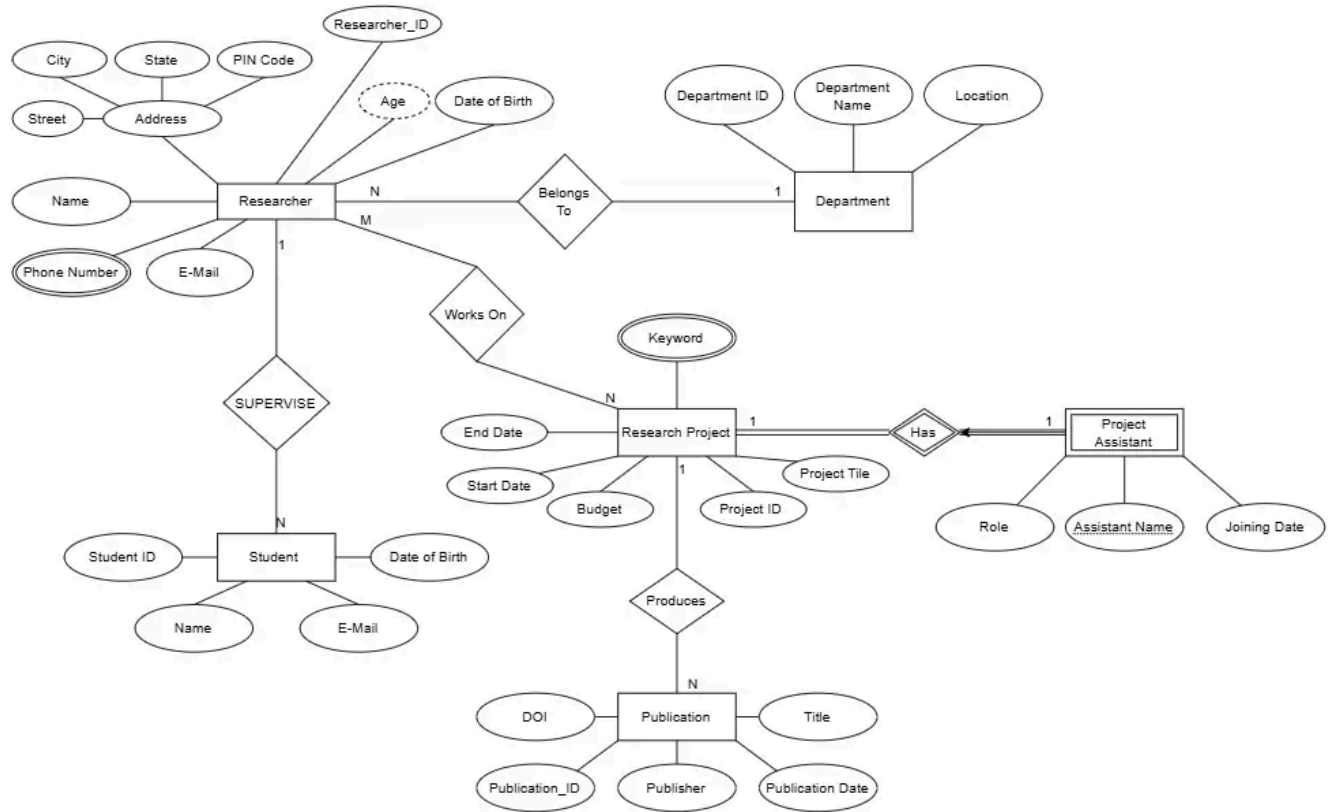
13. Identify appropriate foreign keys after converting the ER model into relations.

Relation	Foreign Key	References
RESEARCHER	Department_ID	DEPARTMENT(Department_ID)
RESEARCHER_PROJECT	Researcher_ID	RESEARCHER(Researcher_ID)
RESEARCHER_PROJECT	Project_ID	RESEARCH_PROJECT(Project_ID)
RESEARCHER_STUDENT	Researcher_ID	RESEARCHER(Researcher_ID)
RESEARCHER_STUDENT	Student_ID	STUDENT(Student_ID)
PUBLICATION	Project_ID	RESEARCH_PROJECT(Project_ID)
PROJECT_ASSISTANT	Project_ID	RESEARCH_PROJECT(Project_ID)

14. Identify all major relationships between the entities.

Relationship	Entities Involved	Cardinality	Meaning
BELONGS	Department ↔ Researcher	1 : N	One department has many researchers; each researcher belongs to one department.
WORKS	Researcher ↔ Research Project	M : N	A researcher can work on many projects, and a project can involve many researchers.
SUPERVISES	Researcher ↔ Student	1 : N	One researcher can supervise many students; each student works under one researcher.
PRODUCES	Research Project ↔ Publication	1 : N	One project can produce many publications; each publication belongs to one project.
HAS_ASSISTANT	Research Project ↔ Project Assistant	1 : N	One project can have many assistants; each assistant belongs to one project. This is an identifying relationship because Project Assistant is a weak entity.

15. Draw the complete ER diagram using standard ER notation.



16. Prepare a table showing each attribute and its attribute type and key status

Entity	Attribute	Attribute Type	Value	Key Status
Researcher	Researcher_ID	Simple	Single	Primary Key
Researcher	Name	Simple	Single	Non-key
Researcher	Date_of_Birth	Simple	Single	Non-key, Stored
Researcher	Age	Simple/Derived	Single	Derived
Researcher	Email	Simple	Single	Non-key

Researcher	Phone_Number	Simple	Multi-valued	Non-key
Researcher	Address	Composite	Single	Non-key
Researcher	Street	Simple component	Single	Non-key
Researcher	City	Simple component	Single	Non-key
Researcher	State	Simple component	Single	Non-key
Researcher	PIN_Code	Simple component	Single	Non-key
Department	Department_ID	Simple	Single	PrimaryKey
Department	Department_Name	Simple	Single	Non-key
Department	Location	Simple	Single	Non-key
Research Project	Project_ID	Simple	Single	Primary Key
Research Project	Project_Title	Simple	Single	Non-key
Research Project	Start_Date	Simple	Single	Stored
Research Project	End_Date	Simple	Single	Stored
Research Project	Budget	Simple	Single	Stored
Research Project	Keyword	Simple	Multi-valued	Non-key
Student	Student_ID	Simple	Single	Primary Key
Student	Name	Simple	Single	Non-key

Student	Email	Simple	Single	Non-key
Student	Date_of_Birth	Simple	Single	Stored
Publication	Publication_ID	Simple	Single	PrimaryKey
Publication	Title	Simple	Single	Non-key
Publication	Publication_Date	Simple	Single	Stored
Publication	DOI	Simple	Single	Non-key
Publication	Publisher	Simple	Single	Non-key
Project Assistant	AssistantName	Simple	Single	Partial Key
Project Assistant	Role	Simple	Single	Non-key